

Nicholas Bockrath

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OBJECTIVE

Electrical Engineering senior seeking an internship/full-time role in power systems, controls, defense applications, or power generation/transmission. Interested in applying circuit analysis, power systems modeling, and embedded systems experience to realworld engineering challenges.

EDUCATION Bachelor of Science in Electrical Engineering

Arizona State University, Phoenix, Arizona Expected
Graduation: May, 2026

- Relevant Coursework:
 - Circuits I & II, Digital and Analog Circuits, Digital Systems and Circuits
 - Signals & Systems I & II, Control Systems, Random Signal Analysis
 - Properties of Electronic Materials (Semiconductor Physics)
 - Electric Machines, Power Electronics, Electrical Power Plants, Nuclear Power Engineering, Power Systems Analysis
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TECHNICAL SKILLS

Software: MATLAB, Simulink, LTspice, Python, C/C++

Hardware: Oscilloscopes, Multimeters, Power Supplies, Soldering

Engineering: Circuit Design, Power Flow Analysis, Motor Modeling, Control Systems, Signal Analysis

Other: Microsoft Office

PROJECTS Motor Equivalent Circuit Analysis

- Modeled induction motor using no-load and blocked-rotor test data
- Calculated parameters (R_1 , X_1 , R_2' , X_2' , X_m) and predicted performance

- Simulated efficiency and torque-speed characteristics in MATLAB **Power Flow**

Study (Load Flow Analysis)

- Performed power flow calculations
- Analyzed voltage regulation and line losses in multi-bus system
- Implemented solution in MATLAB

2.5 MW Pulse Power Supply Prototype (Team Lead)

- Worked with the Los Alamos National Laboratory to build a 2.5 MW modulator prototype designed for an academic environment
- Integrated MOSFETs, isolated gate drivers, and current-sense amplifiers that are programmed via STM32CubeIDE
- The device's adjustable PWM control, current limiting, and real-time waveform visualization provides an educational and safe platform for students to work with in any college laboratory.

ADDITIONAL EXPERIENCE/CREDENTIALS

- Security Clearance: Top Secret (Expired)
- Reference – Jesus Valladares:
 - RF Engineering Team Leader at Los Alamos National Laboratory
 - CAPSTONE Project Mentor
 - (956) 746-7675
 - chuy@lanl.gov
- Car Maintenance Demonstrating Problem-Solving Skills and Good Mechanical Intuition:
 - Replaced O2 Sensors
 - Replaced Timing and Power Steering Belts
 - Replaced Brake Pads & Rotors
 - Replaced Shocks
 - Installed & Integrated New Audio System Into Car's Electrical System